Peter (Pedram) Beigi, PhD Candidate

Autonomous Driving Researcher and Machine Learning Engineer

beigi@gwu.edu LinkedIn GitHub Scholar Personal Website

Based in Santa Clara, CA & Washington, DC
Open to Relocate

PROFESSIONAL SUMMARY

Develops perception, prediction, and decision-making systems for autonomous driving. Experienced in building scalable ML pipelines using large-scale multimodal datasets across simulation and real-world environments.

RESEARCH EXPERIENCE

Research Intern
May 2026 - Aug 2026

University of Illinois Urbana-Champaign — Urbana-Champaign, IL

- Working with *Prof. Talebpour* on developing advanced approaches including foundation models, vision-language architectures, diffusion models, imitation learning, and reinforcement learning, leveraging large-scale driving data.

Autonomous Vehicle & Machine Learning Intern
Sep 2022 - Present

FHWA Turner-Fairbank Research Center (Saxton Lab) — McLean, VA

- Leading development of trajectory generation models for ADAS (using physics-based, DL-based and RL-based frameworks), and integrating AV behavior models into large-scale microsimulation (SUMO) tools (*Repos are private for now*).
- Developed and validated perception (sensor fusion, detection, segmentation) and planning (map generalization, localization) algorithms using LiDAR, radar, and video data for enhanced AV situational awareness and mixed-traffic simulation (*Some repos are public here*).
- Built an end-to-end computer vision pipeline (Faster R-CNN, multi-camera MOT, BEV transformation, recursive state estimation, SIFT feature matching, and annotation/data processing) to extract trajectories and support motion prediction on 11,000+ vehicle-km of AV-human interaction data (*Data is published by USDOT*).

Graduate Researcher
Sep 2021 - Present

GW The George Washington University — Washington, DC

- Developed multi-agent reinforcement learning (MARL) frameworks for AV coordination and negotiation in dynamic partially observable environments using CTDE framework with multi-agent PPO actor-critic models. Applied Bayesian optimization and genetic algorithms for the models' calibration.
- Engineered model observability and traceability by developing Logsim DAGs to capture closed-loop metrics, improving experiment reproducibility for the autonomous stack.
- Led creation of an integrated simulation-AI pipeline combining SUMO, CARLA, and Python APIs for closed-loop decision-making experiments using real-time data (trajectory) streaming using twelve 5G-connected cameras.

WORK EXPERIENCE

Autonomous Vehicle Researcher Intern
Sep 2024 - Dec 2024

m moment.ai (startup under **State Farm Insurance**) — Washington, DC

- Developed multi-modal deep learning time-series models (LSTM/GRU) integrating smartphone and AV sensor data to detect driver behavior and health incident risks, outperforming traditional methods by 9%.
- Developed a real-time NVIDIA Jetson/mobile driver monitoring system using AWS IoT pipelines and Alexa integration for in-vehicle alerts and control.

SELECTED PUBLICATIONS

h-index: 6 / citation: 230

- "Understanding the Interaction of Vehicles within an Intersection: An Optimal Control Approach with Differential Games" — INFORMS (2026) (Oral top 10%)
- "Diffusion Process-Based Model for Network Trajectory Propagation" — IEEE T-ITS (2026)
- "Percolation-Based Cloud Computing Framework for Network Reliability Assessment" — IEEE ICMI (2025)
- "Learning to Negotiate Chaos: Residual Multi-Agent Policies over Behavioral Priors" — ICLR (under review)
- "Deep Reinforcement Learning Under Partial Observability: Co-Simulation and Penetration-Aware Policy Learning for Traffic Control" — IEEE VTC Fall (2026)
- "A Hybrid Genetic Algorithm and Reinforcement Learning Approach for Vulnerable Road User Modeling" — TRR (under review)
- "A Deep Reinforcement Learning Control to Explore Impact of V2X Connectivity on Intersection Performance" — TRR (2025)
- "Third Generation Simulation Dataset" — TRR (2024)

TECHNICAL SKILLS

Programming: Python, ROS, MySQL, Git

Libraries: PyTorch, Diffusers, Transformers,OpenCV, RLlib, Gymnasium, Scikit-learn, Keras, Ultralytics

Cloud Computing Platform: Azure ML

Datasets: Waymo Motion, Waymo Perception, nuScenes, Argoverse 2, TGSIM, HighD, INTERACTION

Tools: CARLA, Unity, Power BI, Tableau, SUMO, VISSIM, Synchro, SIDRA

EDUCATION

The George Washington University
Sep 2021 - Exp: May 2027
PhD in Transportation Engineering (AV Systems & AI) - GPA: 3.91/4.00

Sharif University of Technology
Sep 2016 - Feb 2021
Bachelor of Science in Civil Engineering - GPA: 3.80/4.00

PROFESSIONAL CERTIFICATIONS

Social and Behavioral Research Certification
Issued by: Collaborative Institutional Training Initiative (CITI Program)
Credential ID: 71300833 | Aug 2021 - Aug 2027

RELEVANT COURSEWORK

- UMD: CMSC 422 Machine Learning (audited)
- UMD: CMSC 472 Deep Learning (audited)
- Stanford: CS231n DL for Computer Vision (Online)
- GW: CE 6609 Numerical Methods in Engineering (4/4)
- GW: CE 6800 Data Science (4/4)
- GW: ECE 6130 Big Data and Cloud Computing (4/4)
- UMD: CMSC 426 Computer Vision (audited)
- UMD: CMSC 828 Visual Learning and Recognition (audited)
- Google DeepMind: Introduction to RL (Online)

WORK AUTHORIZATION

No sponsorship required now or in the future.

Available to start immediately. Available with 3 days' notice for positions in the DC and Bay Area. Available with 1 week's notice for positions in other locations.